

The DGLAP content of the two- $\mathfrak{C}$ row

z integrated first — and the splitting function that comes out

$$S \equiv p^+ + k^+, \quad \zeta \equiv \frac{k^+}{S}, \quad \bar{\zeta} = 1 - \zeta = \frac{p^+}{S}, \quad r \equiv y' - y, \quad d = 2 - 2\epsilon$$

Yes — and it comes out exactly. Doing the z integral first and then writing the bracket in the splitting variable $\zeta = k^+/(p^+ + k^+)$, the entire z -integrated structure collapses to

$$\frac{\pi}{k^{+2}} \left\{ \underbrace{C_{UV}(\zeta)}_{P_{gg}/2N_c} \mathcal{L} \mathcal{X} + \zeta \bar{\zeta} [\mathcal{X} - 2\mathcal{R}] \right\}, \quad C_{UV}(\zeta) = \frac{\zeta}{\bar{\zeta}} + \frac{\bar{\zeta}}{\zeta} + \zeta \bar{\zeta}.$$

The coefficient of the collinear logarithm is **precisely the $g \rightarrow gg$ DGLAP splitting function**, with no leftover factors: symbolic difference 0, and the whole expression agrees exactly with the p^+ -variable form established earlier. The non-logarithmic remainder is governed by $\zeta \bar{\zeta}$ — the *regular* piece of P_{gg} alone. This also settles what to do with the $\frac{1}{\epsilon}$: it is the **collinear factorisation pole**, absorbed into the gluon distribution with $\mu \rightarrow \mu_F$ (§6).

1. Which variable is the splitting variable

Read the row as a process. $\mathfrak{A}^\dagger(p^+ + k^+, -y')$ and $\mathfrak{A}(p^+ + k^+, -y)$ create a **single** gluon of light-cone momentum $S = p^+ + k^+$ off the source ρ . The two $\mathfrak{C}$ vertices then split it into the **measured** gluon (k^+) and the **unobserved** one (p^+). Two gluons in the final state, one parent. So the DGLAP variable is the measured daughter's fraction of the parent,

$$\zeta = \frac{k^+}{p^+ + k^+}, \quad \bar{\zeta} = \frac{p^+}{p^+ + k^+}, \quad \zeta + \bar{\zeta} = 1,$$

not $\xi = p^+/k^+$. The two combinations that appear in $\mathfrak{C}_{1jik}$ become

$$a \equiv \frac{S}{p^+} = \frac{1}{\bar{\zeta}}, \quad b \equiv \frac{S}{k^+} = \frac{1}{\zeta},$$

and the net longitudinal factor left by the two $\delta^{(2)}$'s is $\frac{p^+}{k^+ S^2} = \frac{\bar{\zeta}}{\zeta S^2}$.

2. The bracket in ζ

From the index-engine contraction (verified in general d), the three structures carry

$$c_1 = d_\perp - 2a - 2b = d_\perp - \frac{2}{\bar{\zeta}} - \frac{2}{\zeta}, \quad c_2 = a^2 + b^2 = \frac{1}{\bar{\zeta}^2} + \frac{1}{\zeta^2}, \quad c_3 = 2ab = \frac{2}{\zeta \bar{\zeta}},$$

multiplying $\delta_{km}\delta_{k'm'}$, $\delta_{mm'}\delta_{kk'}$, $\delta_{km'}\delta_{k'm}$ respectively.

3. The z integration, done first

All the z dependence is the rank-2 tensor, which in $d = 2 - 2\epsilon$ is

$$T^{mm'} = \int d^d z \frac{(y-z)^m (y'-z)^{m'}}{(y-z)^2 (y'-z)^2} = C(\epsilon) (r^2 \mu^2)^{-\epsilon} \left[\frac{\delta^{mm'}}{2\epsilon} - \hat{r}^m \hat{r}^{m'} \right], \quad r = y' - y,$$

with $C(\epsilon) = \pi^{1-\epsilon} \frac{\Gamma(1+\epsilon)\Gamma(1-\epsilon)^2}{\Gamma(2-2\epsilon)}$. Writing $X^k = \frac{(x-y)^k}{(x-y)^2}$, $X'^{k'} = \frac{(x'-y')^{k'}}{(x'-y')^2}$ and

$$\mathcal{X} = X \cdot X', \quad \mathcal{R} = (\hat{r} \cdot X)(\hat{r} \cdot X'),$$

the three structures contract as

$$\delta_{km}\delta_{k'm'} \rightarrow T^{kk'} X^k X'^{k'}, \quad \delta_{km'}\delta_{k'm} \rightarrow T^{k'k} X^k X'^{k'}, \quad \delta_{mm'}\delta_{kk'} \rightarrow (\text{tr } T) \mathcal{X}.$$

The momentum sum rule kills the $\frac{2}{k^+}$ term. $T^{mm'}$ is symmetric, so the first two contractions are *identical* and only $c_1 + c_3$ matters:

$$c_1 + c_3 = d_\perp - \frac{2}{\bar{\zeta}} - \frac{2}{\zeta} + \frac{2}{\zeta\bar{\zeta}} = d_\perp + \frac{2}{\zeta\bar{\zeta}}[1 - \zeta - \bar{\zeta}] = d_\perp.$$

In the p^+ variables this cancellation looked like an accident. Here it is manifestly the statement $\zeta + \bar{\zeta} = 1$ — the two daughters share the parent's momentum.

4. Expanding in ϵ

With $d_\perp = 2 - 2\epsilon$ kept consistently (CDR = $\overline{\text{MS}}$),

$$\frac{d_\perp}{2\epsilon} = \frac{1}{\epsilon} - 1, \quad \frac{\text{tr } T}{C(\epsilon)(r^2\mu^2)^{-\epsilon}} = \frac{d}{2\epsilon} - 1 = \frac{1}{\epsilon} - 2,$$

and $\frac{C(\epsilon)(r^2\mu^2)^{-\epsilon}}{\epsilon} = \pi[\mathcal{L} + 2]$ in $\overline{\text{MS}}$, with $\mathcal{L} = \frac{1}{\epsilon} - \log(4\pi^2\mu^2r^2)$. So the brace is

$$\underbrace{d_\perp \left[\frac{\mathcal{X}}{2\epsilon} - \mathcal{R} \right]}_{c_1+c_3} + \underbrace{c_2 \left(\frac{1}{\epsilon} - 2 \right) \mathcal{X}}_{c_2} \longrightarrow [(\mathcal{L} + 1)\mathcal{X} - 2\mathcal{R}] + \left(\frac{1}{\zeta^2} + \frac{1}{\bar{\zeta}^2} \right) \mathcal{L} \mathcal{X}.$$

Multiplying by the net longitudinal factor $\frac{\bar{\zeta}}{\zeta S^2}$ and using $S^2 = k^{+2}/\zeta^2$, i.e. $\frac{\bar{\zeta}}{\zeta S^2} = \frac{\zeta\bar{\zeta}}{k^{+2}}$:

$$\frac{\pi}{k^{+2}} \left\{ \underbrace{\frac{\bar{\zeta}}{\zeta} \left[1 + \frac{1}{\zeta^2} + \frac{1}{\bar{\zeta}^2} \right] \zeta^2}_{= C_{\text{UV}}(\zeta)} \mathcal{L} \mathcal{X} + \zeta\bar{\zeta}[\mathcal{X} - 2\mathcal{R}] \right\}$$

Two checks, both exact: the difference from the direct ϵ -expansion is 0, and substituting $\zeta = \frac{k^+}{k^+ + p^+}$ reproduces the p^+ -variable result $\frac{1}{k^+} \left[\frac{\mathcal{L}\mathcal{X}}{p^+} + \frac{p^+}{k^{+2}} \mathcal{L}\mathcal{X} + \frac{p^+}{(k^+ + p^+)^2} ((\mathcal{L} + 1)\mathcal{X} - 2\mathcal{R}) \right]$ with difference 0.

5. That coefficient is P_{gg}

$$C_{\text{UV}}(\zeta) = \frac{\zeta}{\bar{\zeta}} + \frac{\bar{\zeta}}{\zeta} + \zeta\bar{\zeta} = \zeta\bar{\zeta} + \frac{1}{\zeta\bar{\zeta}} - 2 = \frac{P_{gg}(\zeta)}{2N_c}, \quad P_{gg}(\zeta) = 2N_c \left[\frac{\zeta}{\bar{\zeta}} + \frac{\bar{\zeta}}{\zeta} + \zeta\bar{\zeta} \right].$$

The colour reduction of this row supplies exactly one factor N_c (from $U^{d'b}(z)U^{db}(z) = \delta^{d'd}$ then $f^{c'da}f^{cda} = N_c\delta^{c'c}$), so the two combine into

$$N_c C_{\text{UV}}(\zeta) = \frac{P_{gg}(\zeta)}{2}$$

with no spare factors. And the non-logarithmic remainder carries $\zeta\bar{\zeta}$ — the regular, non-singular piece of the splitting function, the one term of P_{gg} that has no pole at either endpoint.

6. What the logarithm and the pole are

$r = y' - y$ is conjugate to the measured transverse momentum: the surviving phase is $e^{-ik \cdot r}$. So $\mathcal{L} = \frac{1}{\epsilon} - \log(4\pi^2\mu^2r^2)$ is the logarithm between the scale μ and the hard transverse scale r^{-1} , which the remaining y, y' Fourier transform ties to $k_\perp$. $P_{gg}(\zeta)$ times a logarithm of that ratio, plus a $\frac{1}{\epsilon}$ pole — that is the DGLAP structure.

This sharpens what I said before. I established that the $\frac{1}{\epsilon}$ is infrared (the $|z| \rightarrow \infty$ end), not a UV renormalisation pole, and said it must cancel against the companion rows. The P_{gg} identification says something better: a **collinear (mass) singularity is infrared in origin**, and it does not have to cancel — it is **absorbed into the gluon distribution** by DGLAP, with μ becoming the factorisation scale μ_F . Both statements hold together:

- **Do not** run α_s with this μ , and do not treat the pole as UV. (unchanged)
- **Do** remove it with the collinear counterterm $-\frac{1}{\epsilon} \frac{P_{gg}}{2} \otimes (\text{LO})$; the coefficient is exactly right for that, which is the evidence that this is a factorisation pole and not an uncancelled divergence.

Worth confirming against the companion rows that nothing is double counted, but the coefficient coming out as P_{gg} with no leftovers is a strong check.

7. The ζ integral

At fixed k^+ , $S = k^+/\zeta$, so

$$dp^+ = \frac{k^+}{\zeta^2} d\zeta, \quad \zeta \in \left[\zeta_- = \frac{k^+}{\vee}, \quad \zeta_+ = \frac{k^+}{k^+ + \Lambda} \right],$$

and $\frac{d\zeta}{\zeta^2} = \frac{1}{\zeta} \frac{dS}{S}$, which is the usual DGLAP convolution measure. The primitive is elementary:

$$\int \frac{d\zeta}{\zeta^2} C_{UV}(\zeta) = F(\zeta), \quad F(\zeta) = 2 \log \zeta - \log(1 - \zeta) - \zeta + \frac{1}{\zeta} - \frac{1}{2\zeta^2} \quad (F' = C_{UV}/\zeta^2 \checkmark),$$

$$\int_{\zeta_-}^{\zeta_+} \frac{d\zeta}{\zeta^2} C_{UV}(\zeta) = \underbrace{\log \frac{k^+}{\Lambda}}_{\text{rapidity}} - \frac{1}{2} + \underbrace{2 \log \frac{\vee}{k^+} - \frac{\vee}{k^+} + \frac{\vee^2}{2k^{+2}}}_{\text{kinematic end}}.$$

k^+	Λ	$\vee$	quadrature	formula
1.0	10^{-6}	50	1221.139354	1221.139557
2.0	10^{-5}	30	114.619847	114.622173
0.5	10^{-7}	80	12665.075276	12665.075296

- $\zeta \rightarrow 1$ is $p^+ \rightarrow 0$: $-\log(1 - \zeta) \rightarrow \log \frac{k^+}{\Lambda} = \ell_k$. This is the rapidity logarithm, coefficient exactly 1 — the piece that goes to JIMWLK, and the piece the $[\frac{1}{p^+}]_+$ prescription isolates.
- $\zeta \rightarrow 0$ is $p^+ \rightarrow \vee$: the unobserved gluon takes essentially all the momentum. The $\frac{1}{\zeta^3}$ there gives the $\frac{\vee^2}{2k^{+2}}$ growth already flagged. This is where the eikonal treatment of $\mathfrak{A}$ has to be cut by real kinematics.

If instead you impose $p^+ \leq k^+$ (the measured gluon the harder one, $\zeta \geq \frac{1}{2}$) the $\vee$ problem vanishes and the integral is clean:

$$\int_{1/2}^{\zeta_+} \frac{d\zeta}{\zeta^2} C_{UV}(\zeta) = \log \frac{k^+}{\Lambda} + \log 2 = \log \frac{2k^+}{\Lambda} \quad (\text{checked to } 3 \times 10^{-10}).$$

And the familiar sum rule is visible in the same function: $\int_0^1 [\zeta \bar{\zeta} + \frac{1}{\zeta \bar{\zeta}} - 2] d\zeta = \frac{1}{6} - 2 + (\text{endpoint logs}) = 2\ell_k - \frac{11}{6}$, which times $2N_c$ is $4N_c \ell_k - \frac{11N_c}{3} = 4N_c \ell_k - b_0$.

8. The result in DGLAP form

$$\left. \frac{d^3 N^{\text{NLO}}}{d^3 k} \right|_{\mathcal{C}} = \frac{1}{(2\pi)^3} \frac{g^4}{4\pi^4} \frac{1}{2k^+} \int_{k^+/\vee}^{k^+/(k^+ + \Lambda)} \frac{d\zeta}{\zeta^2} \left\{ \frac{P_{gg}(\zeta)}{2} \mathcal{L} \mathcal{X} + N_c \zeta \bar{\zeta} [\mathcal{X} - 2\mathcal{R}] \right\}$$

$$\times \int_{x, x', y, y'} e^{-ik \cdot (y' - y)} [U^{e'c}(y') - U^{e'c}(x')] [U^{ec}(y) - U^{ec}(x)] \rho^{e'}(x') \rho^e(x)$$

$$\zeta = \frac{k^+}{p^+ + k^+}, \quad \bar{\zeta} = 1 - \zeta, \quad r = y' - y, \quad \mathcal{L} = \frac{1}{\epsilon} - \log(4\pi^2 \mu^2 r^2), \quad \mathcal{X} = \frac{(x-y) \cdot (x'-y')}{(x-y)^2 (x'-y')^2}, \quad \mathcal{R} = \frac{[r \cdot (x-y)][r \cdot (x'-y')]}{r^2 (x-y)^2 (x'-y')^2}, \quad P_{gg}(\zeta) = 2N_c \left[\frac{\zeta}{\bar{\zeta}} + \frac{\bar{\zeta}}{\zeta} + \zeta \bar{\zeta} \right].$$

Equivalently, with $\frac{d\zeta}{\zeta^2} = \frac{1}{\zeta} \frac{dS}{S}$, the first term is the standard convolution $\int \frac{dS}{S} \frac{P_{gg}(\zeta)}{2\zeta} \otimes (\text{LO transverse structure})$, and the collinear counterterm that removes the $\frac{1}{\epsilon}$ is the matching $-\frac{1}{\epsilon} \frac{P_{gg}}{2} \otimes (\text{LO})$.

zint/src/row8_dglap.py, row8_dglap2.py — every identity and number above, with the cross-check against the earlier p^+ -variable form.